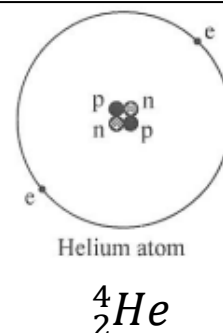


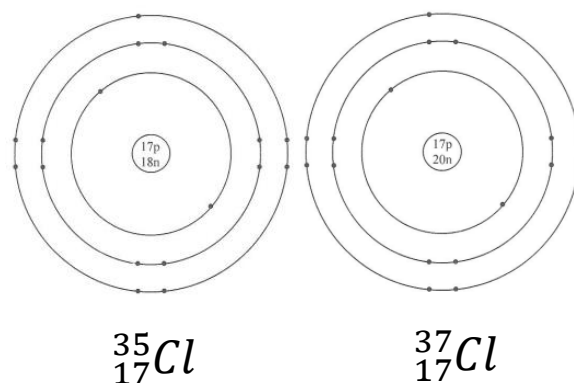
Atomic model

- An atom consists of a nucleus and electrons moving around the nucleus.
- Nucleus contains protons (p) and neutrons (n).
- Symbol of an atom: ${}_Z^AX$
 Z : atomic number = number of protons (p)
 A : mass number = total number of protons + neutrons (p + n)
- All atom must be neutral (number of protons = number of electrons)



Isotopes

- **Isotopes** are different atoms of same elements which have same number of protons but different number of neutrons.
- Isotopes have same atomic number Z but different mass number A .
- Isotopes have **same chemical properties** but **different physical properties**.
- An isotope which is radioactive is called radioisotope.

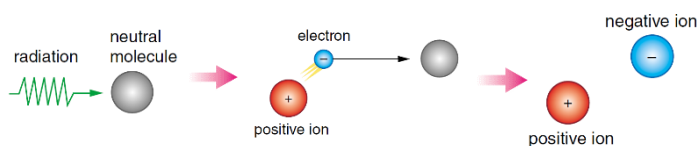


Types of nuclear radiation

	Alpha particles (${}_2^4He$)	Beta particles (${}_{-1}^0e$)	Gamma rays (γ)
Nature	A helium nucleus	A fast moving electrons	Electromagnetic wave
Mass	4 a.m.u.	$\frac{1}{1800}$ a.m.u.	No mass
Charge	2 +	1 -	Neutral
Speed	10 % speed of light (c)	90 % speed of light (c)	speed of light (c)
Range in air	very short range (3-5 cm)	medium range (several meters)	very long range (over 100 m)

(i) Ionizing power: $\alpha > \beta > \gamma$

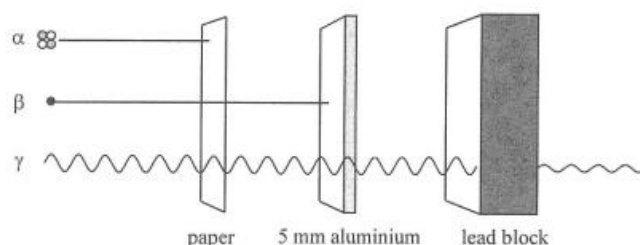
When a radioactive particle passes through air, the particle would collide with the air molecules and knock out an electron from air molecules to produce ion-pairs.



(ii) Penetrating power: $\gamma > \beta > \alpha$

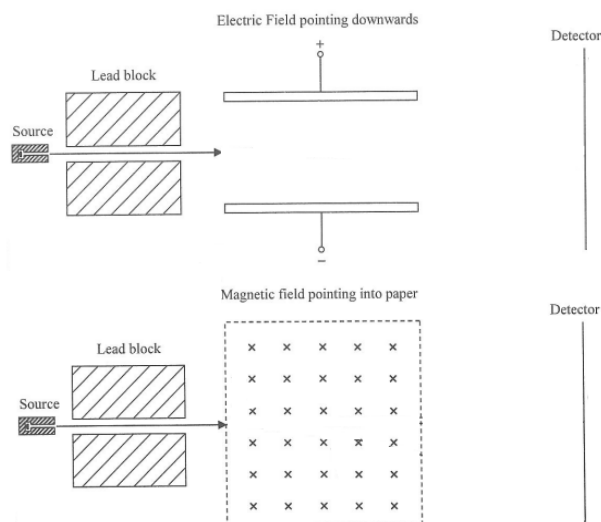
Penetrating power is related to the ionizing power. The greater the ionizing power, the less is the penetrating power.

- **α particles** can be stopped and absorbed by a **piece of paper**. It can also be stopped and absorbed by the **skin** of human body
- **β particles** can be stopped and absorbed by **5 mm thick aluminium block**. It can be stopped and absorbed by the **muscle** of the human body.
- **γ radiation** can be partly absorbed by a lead block. 25 mm thick lead block can reduce **a half of γ radiation**.



(iii) Deflection of radiation in E-field and B-field

- The experiment must be done in vacuum since **α particle** has a very short range in air.
- Only **α** and **β** radiation are deflected.
- The path inside the electric field is **parabolic** while the path inside the magnetic field is **circular**.
- Since **β** is much lighter than **α** , the deflection of **β** is much greater than that of **α** .



Detection of radiation

(i) Photographic film

- When the film is exposed to a radiation, the film would be blackened.
- The more the blackening of the film, the higher is the intensity of the radiation.

Remarks:

A photographic film can detect α radiation but it is wrapped by plastic to prevent being blacken by visible light. Hence, wrapping materials block α radiation.



(ii) GM tube (Geiger-Muller tube)

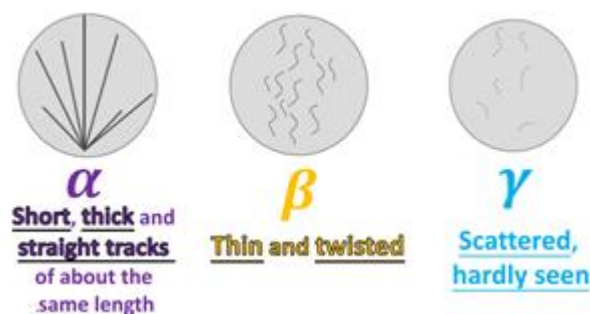
- A GM tube can be connected to a counter to form a GM counter.
- The measured value is called the count rate. Unit of count rate is count per minute (c.p.m.) or counts per second.
- All the 3 types of radioactive radiation can be detected by a GM counter.
- Procedure for using GM counter to measure the range of α in air:
 - (1) Place the alpha source close to and facing the GM counter.
 - (2) Increase their separation gradually and observe the count rate reading.
 - (3) Mark the point for the rapid drop in count rate.
 - (4) Measure the distance between α source and the GM tube with the meter rule to give the range.



Remarks: If no radioactive sample is present, The GM counter can still give a reading due to **background radiation**. Hence, the count rate of a radioactive sample that has deducted the background count rate is called the **corrected count rate**.

(iii) Cloud chamber

- Cloud chamber can detect the tracks of the individual particle
- All the 3 types of radiations can be detected.
- Each ionizing particle leaves a track in the cloud chamber.
- The range and ionizing power of the radiation can be observed from the track.



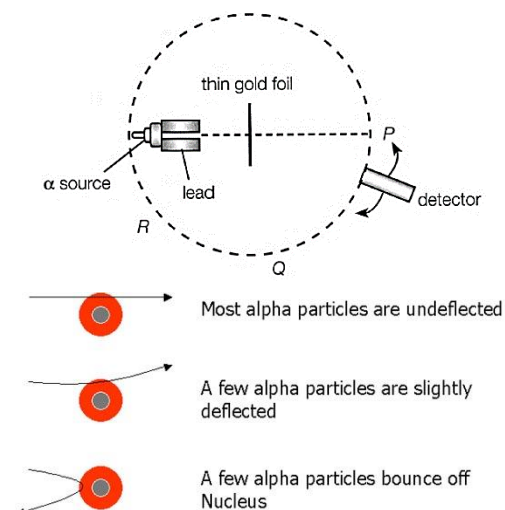
Remarks: If a cloud chamber containing some helium gas is used to observe the tracks of α particles, then a right-angled fork track can be observed. It shows that an α particles and a helium nucleus have the same mass.



Rutherford scattering experiment

The figure shows a simplified set-up to reproduce the experiment of **scattering of α particles by a thin gold foil**, originally carried out by Geiger and Marsden. The detector is moved to various positions and the number of particles detected is measured for the same period of time at each position.

Remarks: the set-up is placed in a **vacuum**.

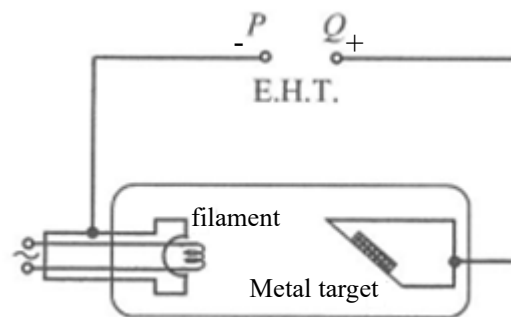


X-rays

● Production of X-ray:

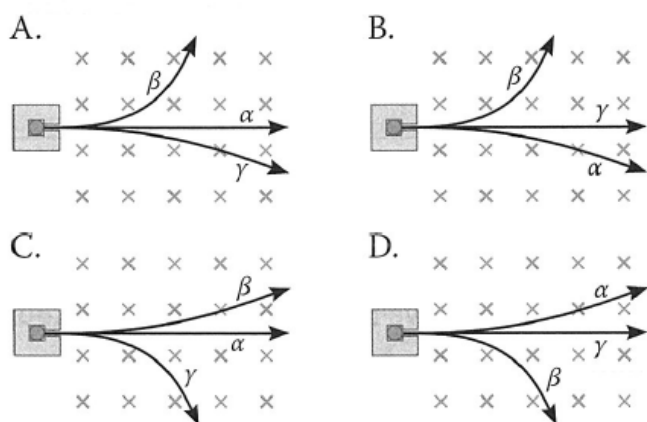
- (i) Electrons are emitted from heated filament.
- (ii) Electrons are then accelerated by high voltage towards the heavy metal.
- (iii) Fast moving electrons hit the heavy metal.
- (iv) X-ray are produced.

- X-ray is an ionizing radiation which has enough high energy to kick out electrons from atoms to become ions and hence it is dangerous.
- X-ray has high penetrating power.
- X-ray can be detected by a film.

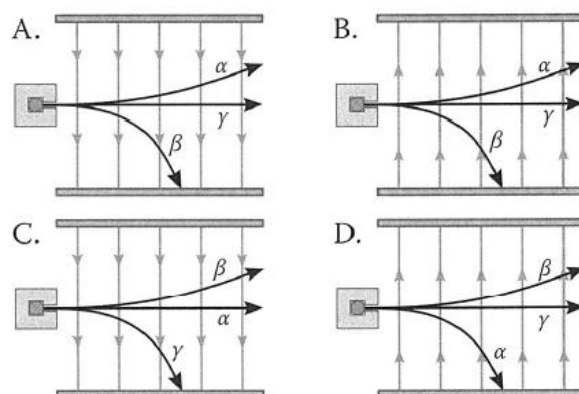


Radiation	Applications
X-rays	- medical diagnosis - inspection of luggage at airports or checkpoints at borders - inspection of trucks at borders - machinery inspection (e.g., jet engines of airplanes) - sterilization of food, mail, etc. - study of crystal structure (crystallography)
α radiation	- used in smoke detectors - installed at the tip of lightning rods
β radiation	- used in thickness gauges in industry to monitor the thickness of products - radiotherapy (healing cancer)
γ radiation	- used in thickness gauges in industry to monitor the thickness of metal sheets - tracers to detect leakage in pipes - medical tracers - sterilization of food, medical equipment and medical waste - radiotherapy (healing cancer)

1. Which of the following diagrams best shows the deflection of α , β and γ radiations in a uniform magnetic field pointing into the paper?

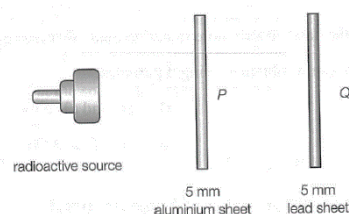


2. A beam consists of α , β and γ radiations is directed into a uniform electric field. Which of the following diagrams correctly shows the deflection of the radiations in the electric field shown?



3. In the figure, the radioactive source emits all types of natural radiations α , β and γ . A radiation detector is placed in region P and then in region Q. What type(s) of radiations is/are detected in each region?

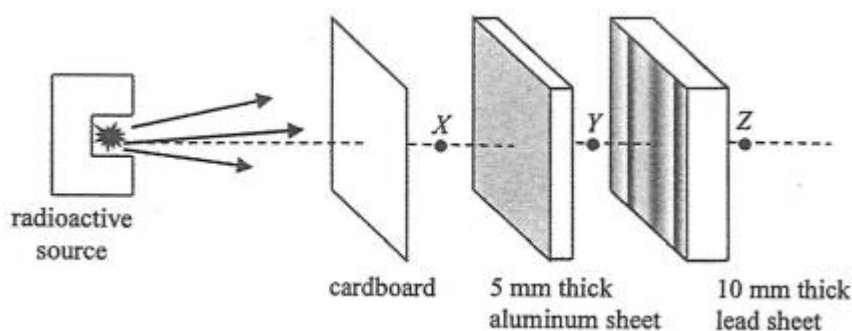
	P	Q
A.	α, β, γ	α, β, γ
B.	β, γ	γ
C.	β, γ	none
D.	γ	γ



4. A radioactive source emits α , β and γ radiations.

Which statement about the radiation(s) detected at positions X, Y, Z indicated in the figure is correct?

- A. No radiation from the radioactive source is detected at Z.
B. Both β and γ radiations can be detected at Y.
C. α radiation can only be detected at X but not at Y and Z.
D. β radiation can only be detected at X but not at Y and Z.



5. Different absorbers are placed in front of a radioactive source in turn to block its radiation. A GM tube is used to detect the residual radiation behind the absorbers. The following table shows the results:

Absorber	Corrected count rate / counts s^{-1}
Nil	300
Paper	298
5 mm aluminium	130
25 mm lead	70
50 mm lead	35

What type(s) of radiation must be emitted by the source?

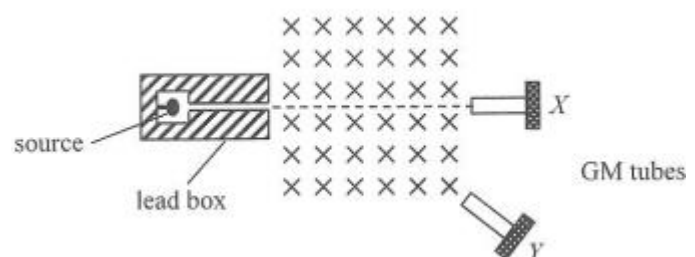
- A. β radiation B. γ radiation
C. β and γ radiations D. α , β and γ radiations
6. Different absorbers are placed in turn between a radioactive source and a Geiger-Muller tube. Three readings are taken for each absorber. The following data are obtained :

Absorber	Count rate / s^{-1}		
—	200	205	198
Paper	197	202	206
5 mm aluminium	112	108	111
25 mm lead	60	62	58
50 mm lead	34	36	34

What type(s) of radiation does the source emit ?

- A. β only
B. γ only
C. β and γ only
D. α , β and γ
7. A radioactive source is placed in front of a uniform magnetic field pointing into the paper as shown above. The count rates recorded by the GM tubes at X and Y are 101 counts per minute and 400 counts per minute respectively. Which of the following deductions must be correct ?

- A. The source does not emit α radiations.
B. The source emits β radiations.
C. The source emits γ radiations.
D. The background count rate is about 100 counts per minute.



8. A radioactive source is suspected to emit α , β and γ . Without other additional apparatuses, which of the following can be used to confirm the identity of the radioactive particles that the source emits?

- (1) Photographic film
(2) Diffusion cloud chamber
(3) Geiger-Muller (GM) counter

- A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only



9. Some factories make use of radioactive source for manufacturing. Workers are required to wear clothes with film badges to measure the dosage of radiation received over a period of time. Which type of radiation below CANNOT be monitored by the film badges?

- A. α -radiation
B. β -radiation
C. γ -radiation
D. X-rays



10. The diagram shows the cloud chamber tracks of a radioactive source.



The source is emitting

- A. α and β only B. α and X – rays only
C. α only D. β only

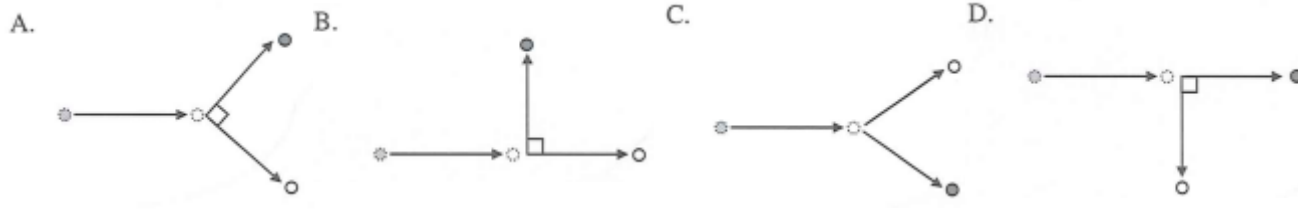


11. Which of the following effects of nuclear radiation is made use of in revealing the paths of radiation in a cloud chamber?

- A. ionization effect
B. magnetic effect
C. chemical effect.
D. heating effect.



12. A stationary helium nucleus is hit by an α particle. Assume the collision is elastic, which of the following shows the possible motions of both particles immediately after the impact?

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13. Right-angled fork tracks of α particles in a cloud chamber with helium nuclei shows that

☐

- A. α particles carry positive charge
- B. α particles have the same mass as helium nuclei
- C. α particles can be deflected by a magnetic field
- D. α particles have a short range in air

14. Which of the following statements about α particles and γ rays is/are correct ?

☐

- (1) They can both be deflected by a magnetic field.
- (2) α particles have a stronger ionizing power than γ rays.
- (3) They are emitted with almost the same speed in radioactive decay.

- A. (1) only
- B. (2) only
- C. (1) & (3) only
- D. (2) & (3) only

15. Which of the following statements about α , β and γ radiations is **incorrect** ?

☐

- A. Only γ radiation can travel through a vacuum.
- B. α radiation can be stopped by an aluminium plate of 5 mm thick.
- C. β particles are fast moving electrons.
- D. γ radiation can blacken a photographic film.

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Which of the following statements about ionizing radiations is/are correct ?

☐

- (1) The ionizing power of α -particles is much stronger than that of β -particles.
- (2) γ -radiation can be completely shielded by a 10 cm thick concrete wall.
- (3) Ionizing radiations α , β and γ all undergo deflection in an electric field.

- A. (1) only
- B. (1) and (2) only
- C. (1) and (3) only
- D. (2) and (3) only

17. Which of the following about the atomic model can be deduced from Rutherford scattering experiment?

- (1) The atom consists of a positively charged nucleus which is very small compared with the atom
- (2) Most of space of the gold atom is empty
- (3) The atom is electrically neutral

- A. (1)and (2)only
- B. (1)and (3)only
- C. (2)and (3)only
- D. (1),(2)and (3)

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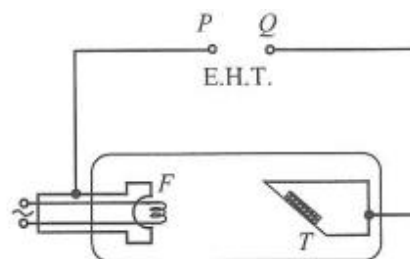
18. Which of the following is not a valid conclusion from Rutherford's alpha particle scattering experiments?

- A. The atom is mainly empty space
- B. The nucleus contains protons and neutrons
- C. The nucleus must be charged
- D. The nucleus must be very small compared to the atom

☐

19. The figure shows a schematic diagram of an X-ray tube in which the filament F and the metal target T are connected to terminals P and Q of an E.H.T. Which statement is correct ?

- A. P is the positive terminal and X-rays are emitted from T .
- B. P is the positive terminal and X-rays are emitted from F .
- C. Q is the positive terminal and X-rays are emitted from T .
- D. Q is the positive terminal and X-rays are emitted from F .


☐

20. Which of the following statements about X-rays is/are correct ?

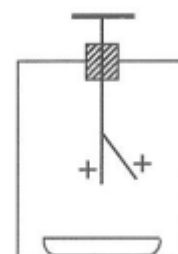
- (1) X-rays consist of fast moving electrons.
- (2) X-rays can blacken photographic films.
- (3) X-rays can be used to detect weapons hidden in luggage.

- A. (1) only
- B. (2) only
- C. (1) & (3) only
- D. (2) & (3) only

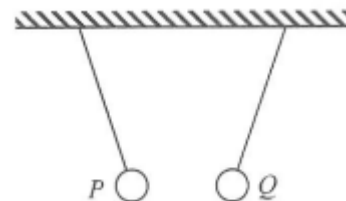
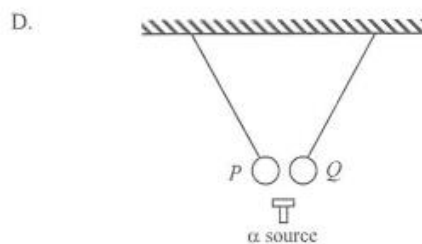
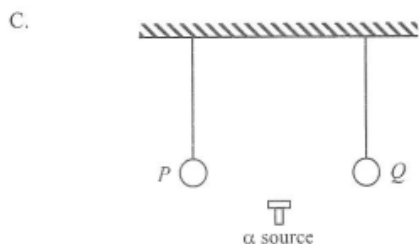
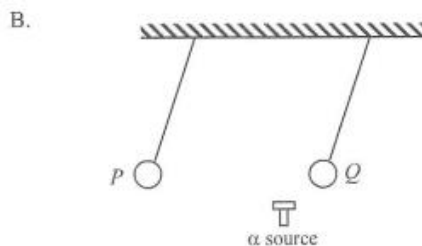
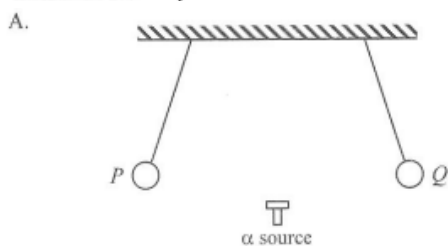
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21. A dish containing a strong α -source is placed inside a gold leaf electroscope containing dry air. If the gold-leaf is originally positively charged, what will happen to it after a few minutes ?

- A. It will increase in divergence.
- B. It will increase in divergence and then decrease.
- C. It will collapse.
- D. It will collapse and then re-diverge.


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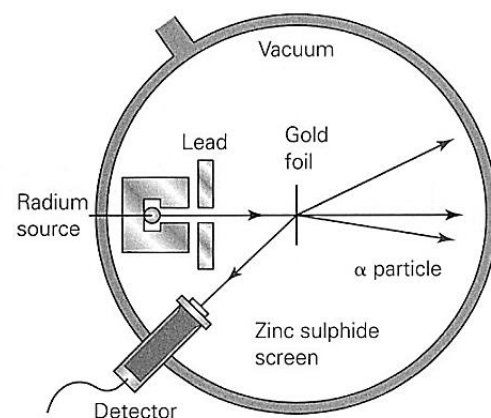
22. In the figure above, two charged metal balls P and Q are hung by insulating threads. P is positively charged while Q is negatively charged. An α source is put near the balls without touching them. Which of the following figures shows the situation after a period of time?



23. The figure shows a simplified set-up of Rutherford's alpha particle scattering experiment. Alpha particles are produced from a radioactive source and emitted through a narrow slit to strike onto a thin gold foil. The detector is a some sort of a movable fluorescent screen.

- (a) Explain (i) why a lead box with a narrow slit was used. (2 marks)
(ii) why the experiment was performed in a vacuum.
- (b) Describe what will be observed as the detector move round the circular track. (3 marks)
- (c) The experiment leads Rutherford to propose a new atomic model at his time. (4 marks)
(i) Describe the existing model at the time of Rutherford.
(ii) Describe Rutherford's atomic model.
- (d) How would the number of alpha particles detected at the region directly opposite the source if the gold foil were about 1 mm thick? (2 marks)

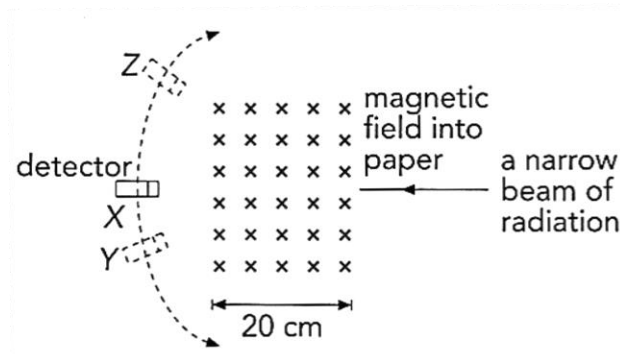
(Total = 11 marks)



24. George carried out an experiment to investigate the deflection of nuclear radiations by a magnetic field. A narrow beam of radiation is directed towards a magnetic field as shown in the figure below in an evacuated chamber.

A radiation detector is moved along the track indicated by the dotted line. It is found that radiation is detected at locations X , Y and Z .

- State the types of radiation detected at locations X , Y and Z . (2 marks)
- What would be the effect, if any, on the readings of the detector at the three locations if the chamber was NOT evacuated? Explain your answer briefly. (4 marks)



25. Figure (b) shows the set-up of an experiment carried out in an evacuated chamber to study the radiations from a radioactive source which undergoes the decays as shown in Figure (a). A magnetic field pointing out of the paper is applied. A detector is moved along the line XY to detect the radiations.

- Sketch and label the paths of α , β and γ radiations emitted from the source. (3 marks)
- Briefly explain
 - the use of lead block with a narrow slit in the set-up, and
 - why the experiment is carried out in an evacuated chamber.

(4 marks)

X-----Y

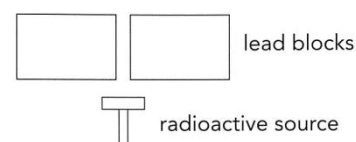


Figure (b)

Radioactive decay

In all radioactive decays, the following two laws must be obeyed:

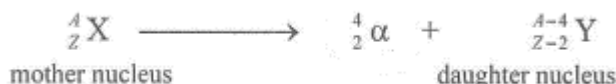
- **Conservation of mass**

The total mass number A must be balanced at both sides in the nuclear equation.

- **Conservation of charge**

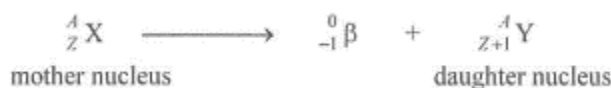
The total atomic number Z must be balanced at both sides in the nuclear equation.

(i) Alpha decay:



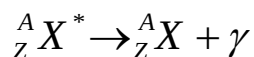
Alpha decay always occurs in a large unstable nucleus. After the decay, the mother nucleus transmutes to the daughter nucleus which is another element. For the new element, its atomic number is **decreased by 2** and its mass number is **decreased by 4**.

(ii) Beta decay:



When a nucleus contains too many neutrons, it may undergo β -decay to form a new element. Compared with the mother nucleus, number of neutrons in the daughter nucleus **decreased by 1** but the number of protons **increased by 1**.

(iii) Gamma emission:

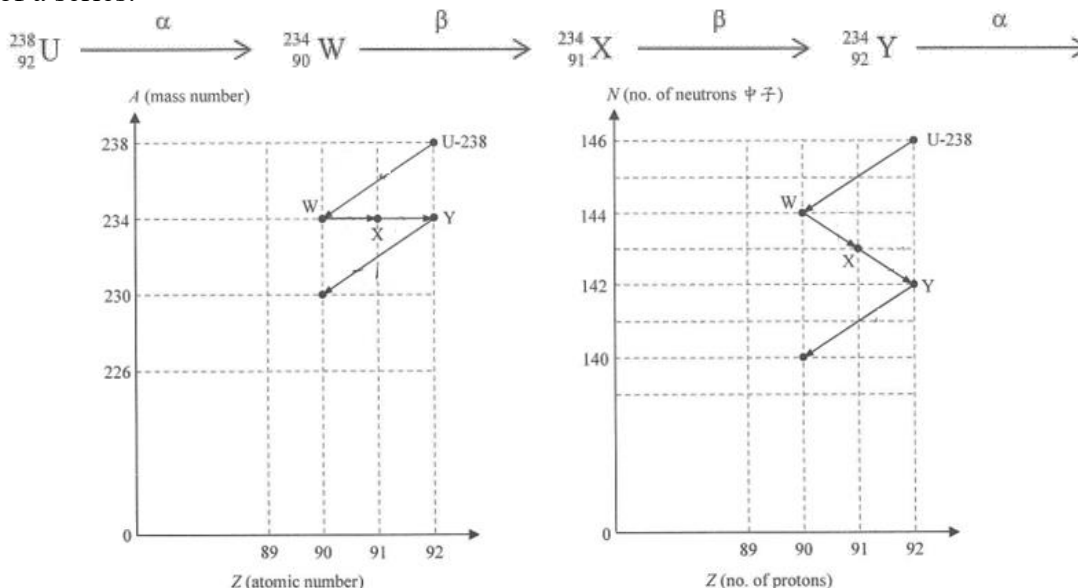


The nucleus releases excess energy in form of γ -radiation. After the γ -emission, there is no change of the mass number and the atomic number of the atom. Also, no new element is formed

Decay series

An **unstable nuclide** can undergo a series of radioactive decays before a **stable daughter nuclide** is formed. This is called **decay series**.

Example of a series:



26. What are the numbers of α and β particles emitted when $^{226}_{88}\text{Ra}$ decays to $^{206}_{82}\text{Pb}$?

	α particles	β particles
A.	4	5
B.	5	4
C.	4	4
D.	5	5

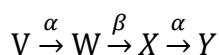
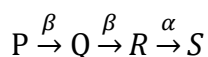
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27. The following equation represents a nuclear fission reaction, producing k neutrons. What are the values of the atomic number z and the number k respectively?

	z	k
A.	36	2
B.	38	2
C.	36	1
D.	38	1

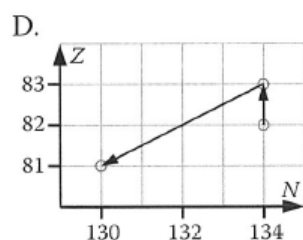
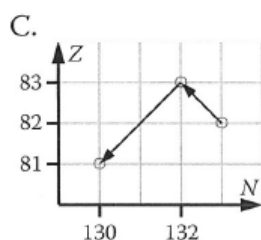
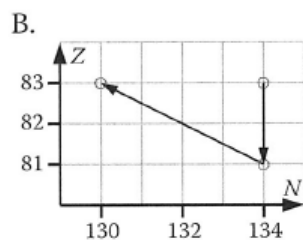
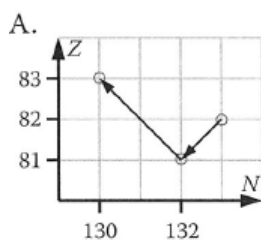

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28. Nuclide P disintegrates to nuclide S and nuclide V disintegrates to nuclide Y as follows:


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If R and V are isotopes, which nuclide(s) is/are the isotope(s) of W?

- A. S only
 B. Y only
 C. P and S only
 D. P and X only
29. Two radioactive decays occur. Which of the following graphs of atomic no. Z against neutron no. N is correct?


☐

30. Which of the following descriptions about the three decays is/are correct?
- (1) Only α decay affects the mass number of the radioactive nuclide.
 - (2) Only β decay is random.
 - (3) Only γ decay has no effect on the atomic number.
- A. (1) only
 B. (2) only
 C. (1) and (3) only
 D. (2) and (3) only

☐

31. Figure (a) shows part of a decay series.

(a) State what particles are emitted in the decay from

(i) C to D, and (1 mark)

(ii) D to E. (1 mark)

(b) The final stable nuclide of the series is lead-207 whose atomic number is 82 and mass number is 207.

Find the total number of

(i) α particles, and (2 marks)

(ii) β particles (2 marks)

Emitted from A to lead-207.

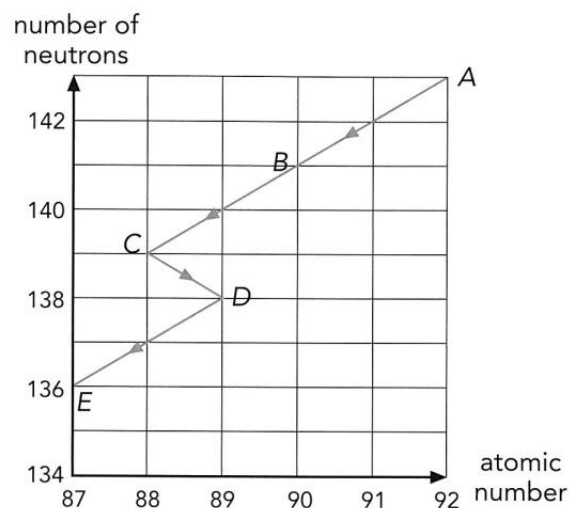
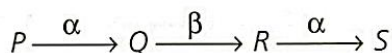
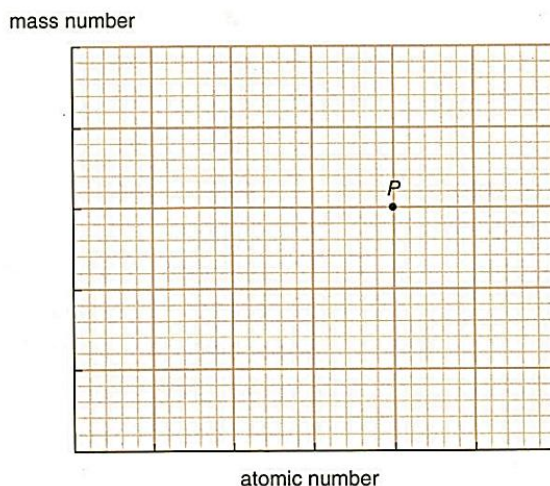


Figure (a)

32. Nuclide *P* has 142 neutrons and 92 protons. It disintegrates to nuclide *S* in the following decay series:



(a) In the figure below, mark the scales of the axes and draw a graph to represent the above decay series. (3 marks)

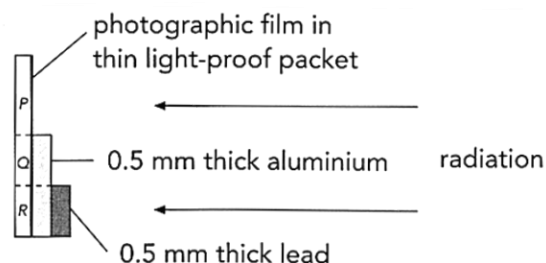


(b) Write a single equation to show how nuclide *P* decays to nuclide *S*.

(1 mark)

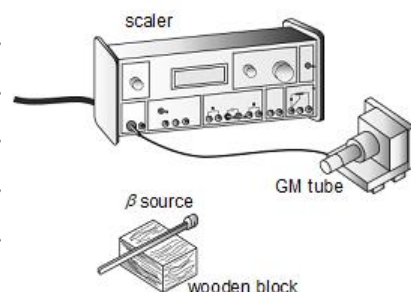
33. The figure below shows one type of badge which is worn by workers at a nuclear power station. The photographic film, which is contained in a thin light-proof packer, turns dark when developed if it has received radiation.

- (a) Describe the appearance of the film when developed if the worker has received an excessive amount of
- β radiation, (2 marks)
 - γ radiation. (1 mark)
- (b) Why is a thin light-proof packet used to surround the photographic film? (1 mark)
- (c) (i) What is the purpose of wearing the film badge? (2 marks)
- (ii) What information can be obtained from the film? (4 marks)



34. A radioactive source is placed in front of a GM tube as shown. Different objects are placed between the source and the GM tube. The count rates measured are shown in the table.

Trial	Object	Count rate / counts per second
1	air	905
2	a piece of paper	711
3	aluminium sheet of thickness 5 cm	713
4	lead block of thickness 25 mm	301



- (a) Does the radioactive source emit
- α radiation?
 - β radiation?
 - γ radiation?
- Briefly explain your answer in each case. (6 marks)
- (b) Explain why the count rate in Trial 2 is different from that in Trial 3. (1 mark)

35.

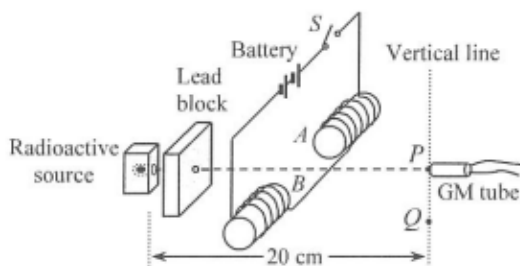


Figure 1

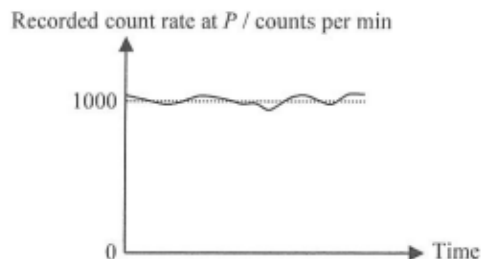


Figure 2

Figure 1 shows a set-up used to study the radiation from a radioactive source. A GM tube is placed at position P , which is at 20 cm from the source. Two coils A and B connected to a battery and a switch S are placed between the source and the GM tube as shown. Initially, S is open and the variation of the count rate recorded by the GM tube with time is shown in Figure 2.

- Explain why the count rate shown in Figure 2 is **not** due to α particles, no matter what kinds of radiation are emitted by the source. (2 marks)
- Now switch S is closed. The GM tube is placed at positions P and Q in turn (see Figure 1) and the count rates recorded are shown in Figure 3 and 4 respectively. When the GM tube is placed at any point vertically above P , an average count rate of 100 counts per minute is recorded at each point.

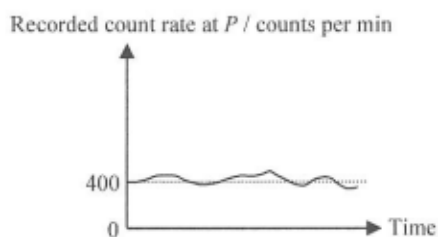


Figure 3

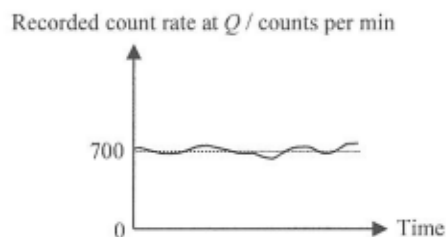


Figure 4

- State the direction of the magnetic field formed between coils A and B . (1 mark)
 - What kind of radiation is recorded when the GM tube is held at any point vertically above P ? Explain your answer. (3 marks)
 - What conclusion about the radiation emitted by the source can you draw from Figure 3 and Figure 4? Explain your answer. (4 marks)
 - Explain why the sum of the average count rates recorded in Figure 3 and Figure 4 is greater than that recorded in Figure 2. (2 marks)
- (c) The above experiment **cannot** determine whether α particles are emitted by the source. Suggest a method for finding out the answer. (2 marks)